



DIGITALIZED COMPREHENSIVE VISUAL PERCEPTION ASSESSMENT- CHILDREN (DCVPA-C)

INSTRUCTION MANUAL

General Description

The Digitalized Comprehensive Visual Perception Assessment- Children (DCVPA-C) is an individually administered assessment designed to evaluate visual perceptual abilities in children aged 4 years 0 months to 12 years 11 months. The assessment includes both static and dynamic visual perception tasks using geometric shapes and familiar everyday objects. It is designed to provide a comprehensive profile of a child's visual perceptual skills across multiple domains in a standardized and engaging digital format.

Purpose

The DCVPA-C is designed to:

- Assess visual perceptual abilities in children aged 4 years 0 months to 12 years 11 months.
- Identify strengths and areas of difficulty across different visual perceptual domains.
- Intended to use in both clinical and research settings by clinicians and educators trained to administer and interpret standardized tests.
- Used as a screening, diagnostic and research purpose.

Examinees

The DCVPA-C is intended for children aged 4 years 0 months to 12 years 11 months.

The assessment may be administered to children with typical development as well as those with developmental, neurological, learning, or visual conditions, provided they are able to understand the task instructions and respond appropriately.

Examiner Qualifications

It is designed primarily for occupational therapists, psychologists, optometrists and other professionals involved in assessing and intervening visual perception skills in children. The examiner should be thorough with the administration and scoring of procedures given in the manual.

Materials Required

1. Laptop or tablet with the DCVPA-C software installed
2. Fully charged device or power supply
3. Stable internet connection
4. Comfortable chair and table
5. Quiet testing room

Testing Time

The DCVPA-C requires 20-30 minutes to administer and about 5 minutes to complete the registration profile for the examinee.

Testing Environment

The testing should be conducted in a quiet, well-lit and ventilated environment free from distractions. Both, the examinee and examiner should be comfortably seated and calm during the test procedure. Visual perception assessments are often conducted during the late mornings to evenings to minimize fatigue and maintain consistent levels of attention and alertness. Cognitive performance, including attention and reaction time, is influenced by

circadian rhythms and varies across the day; therefore, controlling the time of assessment helps improve the reliability of test results (add citation).

<https://www.tandfonline.com/doi/full/10.1080/02643290701754158>

https://www.researchgate.net/publication/332070118_Circadian_Rhythms_in_Attention

Preparation for Testing

- Verify the child's identity.
- Obtain informed consent from the parent or legal guardian.
- Confirm that the child is wearing prescribed spectacles or hearing aids, if applicable.
- Ensure the child is comfortable and attentive.
- Explain that there are no right or wrong feelings about the assessment and encourage the child to try their best.
- Inform the child that they may ask for clarification during the practice items.

Software Navigation

1. Logging In

To begin an assessment, log in to the DCVPA-C platform using your registered email address and password.

2. Home Screen

After logging in, the following options will be available:

- Instruction Manual (PDF) – Opens the administration manual for reference.
- Registration & Assessment– Opens the registration page to enter the examinee's details and begin a new assessment.

- Dashboard – Displays all completed and ongoing assessments.

The Dashboard allows you to:

- View all assessments that have been completed.
- Review individual assessment records.
- Access analytical statistics and summary reports.
- Search and filter assessment records based on the required criteria.
- Download one or more assessment records as PDF reports.

3. Assessment Report

Upon completion of the assessment, a score sheet is generated automatically.

The score sheet can be viewed and downloaded as a PDF immediately after the assessment.

Multiple score sheets can also be downloaded from the Dashboard by selecting or filtering the required assessment records.

INSERT IMAGE HERE FOR REFERENCE

Instruction for Parent/Caregiver/Teacher

- Before beginning the assessment, the examiner will ask a few brief questions regarding the child's demographic information, medical and developmental history, educational background, and any relevant visual or neurological conditions. This information provides important background for interpreting the assessment results and helps ensure an accurate evaluation. Please answer all questions as accurately and completely as possible.
- An electronic informed consent form is included as part of the registration process. Before the assessment begins, the parent, caregiver, or legal guardian must read the consent form, indicate their agreement by selecting the consent checkbox, and submit the form.
- The assessment can only be initiated after informed consent has been provided and the registration has been successfully completed.

Administration and Scoring of DCVPA-C

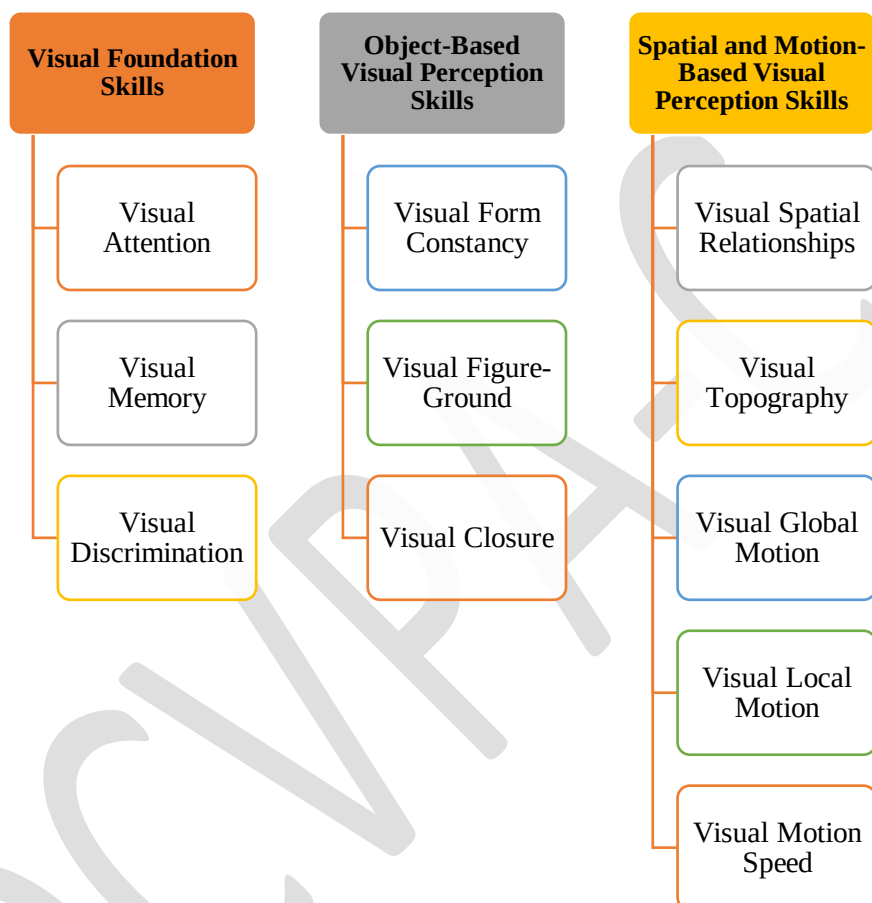
- During the assessment, children were seated approximately 40–60 cm from the screen, which corresponds to the standard near-viewing distance recommended for computer-based visual assessments to ensure clear visibility and reduce visual strain (American Optometric Association, 2020; American Academy of Ophthalmology, 2021). Before beginning the assessment, ensure that the examinee is seated comfortably and can clearly see the screen. Ensure that the examinee is wearing any prescribed spectacles, hearing aids, or other assistive devices before beginning the assessment.

<https://www.aoa.org/healthy-eyes/eye-and-vision-conditions/computer-vision-syndrome>

<https://link.springer.com/article/10.1186/s12886-016-0386-y>

- Administer all items to every examinee, regardless of age.
- The examinee may respond by:
 - tapping or touching the correct option on the screen, or

- pointing to the correct option, or
- verbally stating the correct option (e.g., "1," "2," "3," or "4").
- The assessment consists of 11 visual perceptual domains, grouped into three categories.



- Test items consist of a combination of geometric shapes and common everyday objects.
- Each domain contains 12 items:
 - Items 1 and 2 are trial items and are not scored.

During the trial items, the examiner may repeat the instructions and provide additional prompts to ensure that the examinee understands the task.

- Items 3 to 12 are test items and are scored.

No prompts, demonstrations, feedback, or additional explanations are permitted during the scored test items.

Do not provide feedback such as "correct," "good," or "try again" during the scored items. Neutral encouragement (e.g., "Keep going.") may be provided when necessary without indicating response accuracy.

If the examinee requests clarification during a scored item, repeat the standardized instructions only. Do not explain, demonstrate, or provide hints.

Inform the examinee that there are no penalties for incorrect answers and encourage them to try their best on every item.

- After completing one domain, the next domain will appear automatically. Before beginning each new domain, read the corresponding standardized instructions exactly as written, followed by the trial items and then the scored test items.
- Responses will be recorded immediately after each item according to the scoring guidelines.
- If the assessment is interrupted, resume from the point at which it was paused and document the interruption in the assessment record.
- Short breaks may be provided between domains if the examinee appears tired, distracted, or requires a brief rest. Avoid interrupting a domain once testing has begun unless necessary. If a break is taken, resume the assessment from the point at which it was paused.

Termination Criteria

All items in every domain should be administered to all examinees, regardless of age or performance. There are no basal or ceiling rules, and testing should not be discontinued based on incorrect responses.

Scoring Guidelines

- The computerized system automatically calculates the total number of correct responses, time taken to respond and performance excluding practice items.
- The result is reported as the raw score, time taken, and accuracy. An overall score is obtained for the assessment; along with subtest scores.
- The computerized visual perception assessment generates three types of scores to evaluate the examinee's performance.

1. Raw Score and Total Raw Score

The raw score represents the number of items answered correctly by the examinee. Each correct response receives one point, while incorrect or unanswered items receive zero points. The total raw score is the sum of all correct responses, excluding practice items. This score reflects the examinee's overall accuracy in visual perception tasks.

2. Time Taken and Total Time Taken

The system records the time taken to respond to each item automatically. The total time taken is the sum of the response times for all test items. This score indicates the examinee's processing speed and response efficiency during the assessment.

3. Accuracy Score

The accuracy score combines both correctness and speed. It is calculated using the formula:

$$\text{Accuracy} = (\text{Raw Score} / \text{Time Taken}) \times 100$$

This measure reflects how accurately and efficiently the examinee completes the assessment tasks. A higher accuracy score indicates better performance in terms of both correctness and speed.

Domain-Specific Instructions for DCVPA-C

VISUAL FOUNDATION SKILLS	
Visual Attention	Look at the screen. You will see a group of shapes or objects. Most of them are the same, but “one is different”. Look carefully and find the one that is different from the others. When you find it, tap it on the screen, point to it, or say which one it is.
Visual Memory	Look at the screen. You will see a shape or object for 5 seconds. Look at it carefully and try to remember it. Then, four pictures will appear. Choose the picture that is exactly the same as the one you saw. When you find it, tap it on the screen, point to it, or say which one it is.
Visual Discrimination	Look at the screen. You will see one shape or object at the top of the screen and four shapes or objects below it. Find the one that is exactly the same as the shape or object at the top. When you find it, tap it on the screen, point to it, or say which one it is.
OBJECT-BASED VISUAL PERCEPTION SKILLS	
Visual Form Constancy	Look at the screen. You will see one shape or object at the top of the screen and four shapes or objects at the bottom. Find the shape or object that is the same as the one at the top. It may be bigger or smaller, darker or lighter, or turned sideways or upside down, but it is still the same shape.

	When you find it, tap it on the screen, point to it, or say which one it is.
Visual Figure-Ground	Look at the screen. You will see a shape or object hidden in a busy picture. Look carefully and find the hidden shape or object, even if there are many other colors, patterns, or details around it. When you find it, tap it on the screen, point to it, or say which one it is.
Visual Closure	Look at the screen. You will see a shape or object at the top of the screen that is blurred or only partly shown. Look carefully and find the shape or object below that is exactly the same as the one at the top. When you find it, tap it on the screen, point to it, or say which one it is.
SPACE AND MOTION-BASED VISUAL PERCEPTION SKILLS	
Visual Spatial Relationships	Look at the screen. You will see one shape or object at the top and four shapes or objects at the bottom. Look carefully at the shape at the top. You can imagine turning the shapes in any direction. Three of the shapes are the same as the one at the top, even if they are turned. One shape is different because it is the opposite (a mirror image). Find the shape that is different from the others. When you find it, tap it on the screen, point to it, or say which one it is.
Visual Topography	Look at the screen. You will see a path or route at the top of the screen. Look carefully at the path. Find the same path from the four choices below. The colors may be different, but the path is the same.

	When you find it, tap it on the screen, point to it, or say which one it is.
Visual Global Motion Perception	Look at the screen. You will see a moving pattern or a group of moving objects. Look carefully and find which way the whole pattern is moving. Choose the picture below that shows the same direction. When you find it, tap it on the screen, point to it, or say which one it is.
Visual Local Motion Perception	Look at the screen. You will see a picture or pattern where only part of it is moving. Look carefully and find which way the moving part is going. Choose the correct direction from the four options below. When you find it, tap it on the screen, point to it, or say which one it is.
Visual Motion Speed	Look at the screen. You will see two moving shapes or objects. Look carefully and decide which one is moving faster. When you find it, tap it on the screen, point to it, or say which one it is.

Technical Requirements

- Laptop or tablet with a minimum _____ display.
 - Screen resolution of at least _____ pixels (or equivalent).
 - Stable internet connection.
 - Minimum internet speed: 2 Mbps download and 1 Mbps upload.
 - Updated web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari).
- DCVPA-C is compatible on both, windows and Ios based devices.
- Touchscreen capability is recommended but not mandatory.

Troubleshooting

If technical problems occur:

- verify the internet connection,
- restart the application if necessary,
- resume the assessment from the most recently completed item if supported,
- document any interruptions.

Confidentiality and Ethical Considerations

- All assessment records should be treated as confidential.
- Only authorized personnel should access assessment data.
- Obtain informed consent before administration.